

Question Number	Answer	Mark
5(a)(i)	The only correct answer is B endemic <i>A is not correct because the correct term is endemic</i> <i>C is not correct because the correct term is endemic</i> <i>D is not correct because the correct term is endemic</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	A calculation in which: • correct difference (1) • correct percentage change (1)	<u>Example of calculation:</u> $950 - 7100 = (-)6150$ $(-6150 \div 7100) \times 100 = (-)87(\%)$ Correct answer scores full marks	(2)

Question Number	Answer
*5(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • common ancestor {lived on Hawaii islands / had shorter beak than Amakihi / less powerful beak than Palila} • colonisation of new island / geographical isolation • change in the environment / competition for food / (new) selection pressures • genetic variation in population / mutation resulted in new allele(s) • some alleles conferred an advantage therefore bird more likely to survive and reproduce than other birds and pass on those advantageous alleles to next generation • repeated over many generations leading to new species • idea of reproductive isolation Amakihi • Amakihi has a longer beak • (therefore) was able to drink (more) nectar from flowers / tree sap / access more spiders / insects • Amakihi has larger population due to being adapted to live in more habitats / idea that it can access more types of food sources Palila • Palila has a {more powerful beak / beak which could crack open the coat (of seed)} • Palila was able to eat (more) seeds / berries

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	evolution of generic new species explained with basic information 1 mark – very limited explanation of evolution of a (generic) new species 2 marks – more detailed explanation of evolution of a (generic) new species OR one point about generic explanation plus a basic linkage to either Amakihi or Palila
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Level 1 plus some linkages to either Amakihi or Palila 3 marks = basic linkage for one 4 marks = detailed linkage for one or basic for both
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Level 2 plus detailed linkage to both Amakihi and Palila 5 marks = detailed linkage for one and basic for one 6 marks = detailed linkage for both

Question Number	Answer	Additional guidance	Mark
5(c)	An answer that makes reference to the following points: • molecular phylogeny / analysis of (the sequences in) biological molecules (1) • (therefore) the species with the {most similarities / fewest differences} (are the most closely related) (1)	e.g. DNA, mRNA, proteins	(2)